Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Which statement is correct for hydrogen ?

(A) It has a very high ionisation potential

(B) It is always collected at cathode

(C) It can exist both in $+1$ as well as -1 oxidation state

(D) It has same electronegativity as halogens

Q.2

The adsorption of hydrogen by metals is called

(A) Dehydrogenation

(B) Hydrogenation

(C) Occlusion

(D) Adsorption

Q.3

Hydrogen can be fused to form helium at

(A) high temperature and high pressure

(B) high temperature and low pressure

(C) low temperature and high pressure

(D) low temperature and low pressure

Q.4

Which of the following statements in relation to the hydrogen atom is correct?

(A) 3 s, 3p and 3 d orbitals all have the same energy

(B) 3 s and 3p orbitals are of lower energy than 3 d orbital

(C) 3p orbital is lower in energy than 3 d orbital

(D) 3 s orbital is lower in energy than 3 p orbital

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Q.5

Hydrogen molecules differs from chlorine molecule in the following respect

(A) Hydrogen molecule is non-polar but chlorine molecule is polar

(B) Hydrogen molecule is polar while chlorine molecule is non-polar

(C) Hydrogen molecule can form intermolecular hydrogen bonds but chlorine molecule does not

(D) Hydrogen molecule cannot participate in coordination bond formation but chlorine molecule can

Q.6

Very pure hydrogen (99.9) can be made by which of the following processes ?

(A) Reaction of methane with steam

(B) Mixing natural hydrocarbons of high molecular weight

(C) Electrolysis of water

(D) Reaction of salts like hydrides with water

Q.7

Which statement is wrong?

(A) Ordinary hydrogen is an equilibrium mixture of ortho and para hydrogen

(B) In ortho hydrogen spin of two nuclei is in same direction

(C) Ortho and para forms do not resemble in their chemical properties

(D) In para hydrogen spin of two nuclei is in opposite direction.

Q.8

When electric current is passed through an ionic hydride in molten state :

(A) hydrogen is liberated at anode

(B) hydrogen is liberated at cathode

(C) hydride ion migrates towards cathode

(D) hydride ion remains is solution

Q.9

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When two ice cubes are pressed over each other, they unite to form one cube. Which of the following forces is responsible to hold them together?

(A) Hydrogen bond formation

(B) van der Waals forces

(C) Covalent attraction

(D) Ionic interaction

Q.10

When a substance Areacts with water it produces a combustible gas B and a solution of substance C in water. When another substance D reacts with this solution of C, it also produces the same gas B on warming but D can produce gas B on reaction with dilute sulphuric acid at room temperature. A imparts a deep golden yellow colour to a smokeless flame of Bunsen burner. A, B, C and D respectively are

(A) $\mathrm{Na}, \mathrm{H_2}, \mathrm{NaOH}, \mathrm{Zn}$

(B) $\mathrm{K}, \mathrm{H_2}, \mathrm{KOH}, \mathrm{Al}$

(C) $\mathrm{Ca}, \mathrm{H_2}, \mathrm{Ca(OH)_2}, \mathrm{Sn}$

(D) $\mathrm{CaC_2}, \mathrm{C_2H_2}, \mathrm{Ca(OH)_2}, \mathrm{Fe}$

Q.11

Which of the following metal evolves hydrogen on reaction with cold dilute $\mathrm{HNO_3}$?

(A) Mg

(B) Al

(C) Fe

(D) Cu

Q.12

The hydride ion $\mathrm{H^-}$ is stronger base than its hydroxide ion $\mathrm{OH^-}$. Which of the following reactions will occur if sodium hydride (NaH) is dissolved in water?

(A) $\mathrm{H^-(aq) + H_2O \rightarrow H_3O^-}$

(B) $\mathrm{H^-(aq) + H_2O(l) \rightarrow OH^- + H_2}$

(C) $\mathrm{H^- + H_2O \rightarrow}$ No reaction

(D) None of these

Q.13

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In context with the industrial preparation of hydrogen from water gas $(CO + H_2)$, which of the following is the correct statement?

(A) CO and H_2, are fractionally separated using differences in their densities

(B) CO is removed by absorption in aqueous Cu_2Cl_2 solution

(C) H_2 is removed through occlusion with Pd

(D) CO is oxidised to CO_2 with steam in the presence of a catalyst followed by absorption of of CO_2 in alkali

Q.14

$2\ g$ of aluminium is treated separately with excess of dilute H_2SO_4 and excess of $NaOH$. The ratio of the volumes of hydrogen evolved is

(A) $2 : 3$

(B) $1 : 1$

(C) $2 : 1$

(D) $1 : 2$

Q.15

When same amount of zinc is treated separately with excess of sulphuric acid and excess of sodium hydroxide solution the ratio of volumes of hydrogen evolved is

(A) $1 : 1$

(B) $1 : 2$

(C) $2 : 1$

(D) $9 : 4$

Q.16

Which physical property of dihydrogen is wrong?

(A) Odourless gas

(B) Tasteless gas

(C) Colourless gas

(D) Non-inflammable gas

Q.17

Which of the following reactions increases production of dihydrogen from synthesis gas?

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(A) $CH_4(g) + H_2O(g) \overset{1270\ K}{\underset{Ni}{\rightarrow}} CO(g) + 3H_2(\ g)$

(B) $C(s) + H_2O(g) \overset{1270\ K}{\rightarrow} CO(g) + H_2(\ g)$

(C) $CO(g) + H_2O(g) \underset{Catalyst}{\overset{673\ K}{\longrightarrow}} CO_2(g) + H_2(g)$

(D) $C_2H_6 + 2H_2O \underset{Ni}{\overset{1270\ K}{\longrightarrow}} 2CO + 5H_2$

Q.18

In which of the following reactions, hydrogen will not be liberated?

(A) Allowing a solution of sodium in liquid ammonia to stand.

(B) Reaction of zinc with aqueous alkali.

(C) Reaction of lithium hydride with B_2H_6.

(D) Electrolysis of acidified water using Pt electrodes.

Q.19

Hydrogen gas is not obtained when zinc reacts with :

(A) Cold water

(B) dil. H_2SO_4

(C) dil.HCl

(D) hot $20\%NaOH$ solution

Q.20

Identify the chemical reaction involved in water gas shift reaction:

(A) $C + H_2O \underset{Ni}{\overset{\Delta}{\rightarrow}} CO + H_2$

(B) $CO + H_2O \underset{Catalyst}{\overset{\Delta}{\longrightarrow}} CO_2 + H_2$

(C) $C_nH_{2n+2} + nH_2\,O \underset{Ni}{\overset{\Delta}{\rightarrow}} nCO + \left(2n + 1\right)H_2$

(D) $C + O_2 \longrightarrow CO_2(g)$

Answers & Solutions

Q.1 Answer:

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It can exist both in $+1$ as well as -1 oxidation state

Solution:

The correct option is . In H_2O, hydrogen has $+1$ oxidation no. In CaH_2, hydrogen has -1 oxidation no.

Q.2 Answer:

Occlusion

Solution:

Occlusion is the phenomenon of adsorption of hydrogen by metal.

Q.3 Answer:

high temperature and high pressure

Solution:

A fusion reaction between hydrogen nuclei is difficult because positively charged nuclei repel each other. However, at very high temperatures of the order of 10^6 to 10^7 K, the nuclei may have sufficient energy to overcome the repulsive forces and thus fuse. This is why, fusion reactions are also called thermonuclear reactions.

Q.4 Answer:

$3\ s, 3p$ and $3\ d$ orbitals all have the same energy

Solution:

In one electron species, such as H -atom, the energy of orbital depends only on the principal quantum number,

n.

Hence answer

i.e. is $< 2\ s = 2p < 3\ s = 3p = 3\ d < 4\ s$

$= 4p \Rightarrow 4\ d = 4\ g$

Q.5 Answer:

Hydrogen molecule cannot participate in coordination bond formation but chlorine molecule can

Solution:

Chlorine has a lone pair of electrons and vacant d-orbitals, thus they can take part in coordinate covalent bond. Hydrogen molecule does not have a lone pair of electrons. So it will not take part in the coordinate bond Both chlorine and hydrogen gas are non-polar molecules as they are bonded to the same element and same electronegative elements.

Q.6 Answer:

Electrolysis of water

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Solution:

Highly pure hydrogen can be obtained by electrolysis of water.

It is the decomposition of water into $O_2 \& H_2$ gas by passing electric current.

At anode, $6H_2O(l) \longrightarrow O_2(g) + 4H_3O^+ + 4e^-$

cathode, $4e^- + 4H_2O(l) \longrightarrow 2H_2(g) + 4OH^-$ **(aq)**

Overall $\longrightarrow 2H_2O(l) \longrightarrow 2H_2(g) + O_2(g)\%$

Q.7 Answer:

Ortho and para forms do not resemble in their chemical properties

Solution:

Ortho and para forms of hydrogen resemble in their chemical properties.

Q.8 Answer:

hydrogen is liberated at anode

Solution:

$$\text{At anode} : 2H^- \longrightarrow H_2 \uparrow + 2e^-$$

Q.9 Answer:

Hydrogen bond formation

Solution:

Two ice cubes stick to each other due to H-bonding

Q.10 Answer:

$Na, H_2, NaOH, Zn$

Solution:

$$2Na + \underset{\text{'A'}}{2H_2O} \rightarrow \underset{\text{C'}}{2NaOH} + H_2 \uparrow$$

$$Zn + \underset{\text{'D'}}{2NaOH} \rightarrow Na_2ZnO_2 + \underset{\text{' C'}}{H_2} \uparrow$$

$$Zn + \underset{\text{'D'}}{\text{dil. } H_2SO_4} \rightarrow ZnSO_4 + \underset{\text{'B'}}{H_2} \uparrow$$

Na produces golden yellow colour with smokeless flame of Bunsen burner

Q.11 Answer:

Mg

Solution:

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$Mg + 2HNO_3(\text{ dil. }) \rightarrow Mg(NO_3)_2 + H_2$
(Mg and Mn give H_2 with $dilHNO_3$)

Q.12 Answer:

$H^-(aq) + H_2O(l) \rightarrow OH^- + H_2$

Solution:

$H^-(aq) + H_2O(l) \rightarrow OH^- + H_2$. Since H^- is a strong base it will abstract H^+ to form H_2O

Q.13 Answer:

CO is oxidised to CO_2 with steam in the presence of a catalyst followed by absorption of of CO_2 in alkali

Solution:

On the industrial scale hydrogen is prepared from water gas according to following reaction sequence

$$\underbrace{CO + H_2}_{\text{water gas}} + \underbrace{H_2O}_{(\text{steam})} \xrightarrow{\text{catalyst}} CO_2 + 2H_2$$

$$\underset{(\text{alkali})}{2NaOH} \longrightarrow Na_2CO_3 + H_2O$$

From the above it is clear that CO is first oxidised to CO_2 which is then absorbed in $NaOH$.

Q.14 Answer:

$1:1$

Solution:

$2Al + 3H_2SO_4 \rightarrow Al_2(SO_4)_3 + 3H_2$
$2Al + 2NaOH + 2H_2O \rightarrow 2NaAlO_2 + 3H_2$
The ratio of volumes of hydrogen evolved is $1:1$.

Q.15 Answer:

$1:1$

Solution:

$Zn + H_2SO_4 \rightarrow ZnSO_4 + H_2$
$Zn + 2NaOH \rightarrow Na_2ZnO_2 + H_2$
$\therefore$ Ratio of volumes of H_2 evolved is $1:1$

Q.16 Answer:

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Non-inflammable gas

Solution:

Hydrogen is a highly inflammable gas. This is because it burns in air to give water.

The reaction is as follows:

$2H_2 + O_2 \rightarrow 2H_2O$

Q.17 Answer:

$CO(g) + H_2O(g) \xrightarrow[\text{Catalyst}]{673\ K} CO_2(g) + H_2(g)$

Solution:

The production of dihydrogen can be increased by reacting carbon monoxide of syngas mixture with steam in the presence of iron chromate as catalyst. $CO(g) + H_2O(g) \xrightarrow[\text{Catalyst}]{673\ K} CO_2(g) + H_2(g)$

This is called water-gas shift reaction

Q.18 Answer:

Reaction of lithium hydride with B_2H_6.

Solution:

$B_2H_6 + 2LiH \rightarrow 2LiBH_4$

Q.19 Answer:

Cold water

Solution:

Zinc is a low reactive metal which does not react with cold water and hence no gas is evolved. However, Zn **reacts with dil.** HCl**, dil.** H_2SO_4 **and hot** $NaOH$ **solution.**

Q.20 Answer:

$CO + H_2O \xrightarrow[\text{Catalyst}]{\Delta} CO_2 + H_2$

Solution:

Fact

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